

CDW68 final replication report: IEEE 68-bus confirmatory replication

CDW68 replication (branch research/cdw-ieee68-replication)

2026-09-13

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- Branch: research/cdw-ieee68-replication, from e03a5312.
- Preregistration: commit 6f188531 (2026-09-13T09:19). Every section below cites the result file it reads.
- Scope: phasor-domain small-signal analysis. No EMT claim is made.

1 Executive decision

CASE B, by the exhaustive preregistered rule (prereg §17): the ranking method generalizes and the reversal does not. The gates came out as follows.

component	gate	result
REV68A	level-D nested reversal in ≥ 50 % of base-stable policies, Model A REAL	FAIL: 0/16 (Clopper–Pearson 95 % upper bound 0.21)
REV68B	the same gate on Model B	FAIL: 0/16; level T also 0/16, so not PARTIAL
RANK68A	R12 gate, Model A	PASS: median ρ 0.996, advantage 0.80, top-5 1.0
RANK68B	R12 gate, Model B	PASS: median ρ 0.981, advantage 0.55, top-5 0.8

Consequence. The paper pivots to portfolio-conditioned dynamic reinforcement ranking. The contextual reversal is reported as IEEE-39 benchmark-specific evidence.

The caveat that travels with the pass. On IEEE-68 the preregistered ranking target is the rightmost transverse eigenvalue.

- In every portfolio of both models it is a slow real control pole: the PSS washout at -0.067 s^{-1} .
- Branch reinforcements move it by at most $1.4\text{e-}3 \text{ s}^{-1}$ (A) and $3.4\text{e-}5 \text{ s}^{-1}$ (B), well below $\tau = 0.01$.
- The electromechanical version of the ranking was defined after the fact. It passes as well (A: ρ 0.998, top-5 1.0) with material effects of up to 0.047 s^{-1} , but it cannot carry the preregistered verdict.

2 Why replication was required

The completed hardening review left one experiment blocking a claims paper: every CDW result came from the IEEE 39-bus system with $D = 0$, no governors and constant-power loads. The review asked two questions:

- whether the nested same-mode reversal (17/19 base-stable holdout policies on IEEE-39) and the portfolio-conditioned branch ranking survive a different network with documented primary-frequency dynamics;
- which of them survive a second converter model.

This campaign is that single test. No other network was searched.

3 IEEE-68 provenance

Full detail: docs/CDW68_MODEL_AUDIT.md and results/CDW68_MODEL_MANIFEST.csv.

- Network (Singh & Pal 2013, IEEE PES TF v3.3):
 - 68 buses, 83 branch rows, 16 of them transformers with taps; 100 MVA, 60 Hz.
 - Transcribed at Gate 3 (commit d9fa097e) and reproduced here: power flow within $5\text{e-}5$ pu and $5\text{e-}5$ deg; all 15 EM modes within $5\text{e-}4$ Hz; Ybus rebuild error 0.0.
- Machines: 16 sub-transient machines.
 - Excitation: DC4B (G1–G8, G10–G12), ST1A (G9), manual (G13–G16).
 - PSS: speed washout plus three lead-lags on G1–G12.
 - Loads: constant impedance. Slack: G16.
- Primary frequency (not part of the published benchmark):
 - PST data16m.m (GridSTAGE copy, sha256 c16e78f4...) documents machine ratings (300–1900 MVA; equivalents 10–12 GVA), rotor damping d_o (zero on G1–G12, equal to H on G13–G16) and a PST tg model-1 governor block that is commented out in the file.
 - PSTess tg.m (sha256 c1584021...) gives the equations.
 - Frozen in inputs/pst_primary_frequency_v1.json.

4 Dynamic model realism

variant	governors	rotor damping	base $\alpha_{\perp}$ (s^{-1})	rightmost EM mode
REAL (confirmatory)	PST tg model 1 (1/R = 25 on rating, Ts 0.1, Tc 0.5, T3 0, T4 1.25, T5 5 s) on every surviving machine	PST d_o	-0.0669 (PSS washout, real)	-0.367 s^{-1} at 0.555 Hz
NOGOV (ablation B)	none	PST d_o	-0.0461 (common-frequency mode, 0.002 Hz)	-0.371 at 0.518 Hz
SP33 (ablation C)	none	D = 0 (published)	-0.0556	-0.118 at 0.520 Hz

- Common to all variants: equilibrium residual 7.3e-13; identical network voltages (difference 1.1e-16); smallest documented-limit margin 3.00.
- Damping effect. The equivalents' documented damping moves the 0.52 Hz inter-area mode from -0.118 to -0.37 s^{-1} .
- Consequence for the tests. With it, the rightmost transverse eigenvalue is a slow real pole in every portfolio. This was established in R1 before any converter portfolio was evaluated, and it is the reason the EM-tracked level T was preregistered next to level D.

5 Converter models

- Model A: the frozen TX4 GFL (69f200df; device sha256 bcf0c76b...).
 - SRF-PLL 53/1400; power filter 0.03 s; outer PI 0.2/8; inner PI 0.25/6 behind x_f 0.15, r_f 0.01.
 - Leaky Q/V regulator with gain g (leak 0.05 rad/s).
 - Rating |S_gen|/0.8; matched dispatch.
 - No current limits, DC link or delays.
- Model B: the WECC library chain TX3-GFL-0.1 (sha256 f07a6a40...).
 - Blocks: PLL2, BusFreq, REGCP1, REECB1, REPCA1; constant Q; forced-PQ equilibrium.
 - Runs in ANDES 2.0.0 on a new transcription SG68D/S/M of the Singh & Pal machine with the PST governor and damping (code/andes68/cdw68_models.py).
 - Not retuned.
- Model B qualification (results/b68/B68_qualification*.json), all criteria met before any B portfolio was run:

check	result
Q1	max $ \Delta\alpha_{\perp} $ against Model A's all-SG base: 9.9e-11 s^{-1} over 16 k values, same status; EM top 3.1e-7
Q2	initialization residual $\leq 1.7e-13$
Q3	descriptor QZ against reduced spectrum 7.7e-10 relative
Q4	no active limiter

- Grid-forming model: none is validated in the repository, so none was run.

6 Preregistration

Files: docs/CDW68_PREREG_V1.md, docs/CDW68_STATISTICAL_PLAN.md and docs/CDW68_CLAIM_MATRIX_V1.csv, committed at 6f188531 before any converter portfolio.

Frozen:

- the candidate set, the 16 policies, the draws and the TDS disturbance;
- $\tau = 0.01 s^{-1}$;
- the level definitions A–D and T;
- the R11/R12 gates and the exhaustive CASE rule.

Deviations (docs/CDW68_DEVIATIONS.md, 15 entries):

- execution fixes (Model B check, JSON, ANDES shards, runtime registration, an R15 empty-store guard);

- two post-hoc analyses (EM-tracked branch ranking; exploratory TDS);
- a correction of an interpretation note (entry 11: NOGOV has three global marginals $\geq \tau$);
- layout-only figure fixes;
- paper and report production after the CASE decision (entries 13–14) and a runtime note (entry 15).

No gate, threshold, candidate, policy or model parameter was changed.

7 Candidate selection

V68 = {G3, G4, G6, G9, G11, G12}. These are the six physical plants with the largest scheduled active power (6.32–13.50 pu), with no tie. The rule extends the Gate 3 rule to both areas and uses no dynamic quantity.

- The area equivalents G13–G16 are excluded.
- Size: $2^6 = 64$ portfolios.
- Replacement: full, with matched dispatch.

The set was not changed after results.

8 Policy design

- Model A: 16 maximin-LHS policies, P68_01–16 (seed 20260913, best of 2000 designs, minimum distance 0.1748).
 - Window: $u \in [0, 1]$ with $g = u^2$, and $k \in [0.5, 2.0]$.
 - P68_01–04 are discovery and are used only to select the static comparator. P68_05–16 are holdout.
- Model B: 16 conditions B68_01–16, carrying the k of the matching A policy.
- All 32 conditions have a stable all-SG base, so none was excluded. No policy search took place.

9 Portfolio census

Output: results/CDW68_PORTFOLIOS.parquet (A and B).

set	portfolios	status	$\alpha_{\perp}$ range (s^{-1})	$\alpha_{\perp}$ in the EM band
A REAL, 16×64	1024	all STABLE	–0.0671 to –0.0581	0 %
A NOGOV, 6×64	384	all STABLE	–0.0612 to –0.0461	0 %
A SP33, 6×64	384	all STABLE	–0.0622 to –0.0539	0 %
B REAL, 16×64	1024	all STABLE	–0.06695 to –0.06688	0 %
A draws, 40×64	2560	all STABLE	max –0.0669	0 %
B draws, 20×64	1280	all STABLE	max –0.0669	0 %

- No portfolio is unstable, so H_{RHP} is empty and κ is undefined in every condition.
- Rightmost EM-band mode:
 - A REAL: 0.550–0.806 Hz (median 0.557 Hz), real part –0.373 to –0.263 s^{-1} ;
 - B: 0.552–0.635 Hz, real part –0.369 to –0.251 s^{-1} .

10 Contextual reversal

Primary result (results/CDW68_R07_A_gate.json, results/CDW68_R07_B_gate.json).

- Level-D nested reversal coverage is 0/16 for Model A and 0/16 for Model B (95 % Clopper–Pearson upper bound 0.21 each).
- Levels A, B and C are also 0/16, at every τ from 0.01 to 0.05.
- The median-magnitude clause is undefined, because no level-D pair exists.

Why levels A–D are empty. The global marginals move a slow real pole.

- The largest level-A $|\Delta\alpha_{\perp}|$ is $8.7e-3 s^{-1}$ (A REAL, P68_16, replacing G12). For B it is $2.3e-5$.
- No global marginal is material, so no sign class is non-zero.

Preregistered secondary, level T (tracked rightmost EM mode): 0/16 for both models.

model	EM-tracked marginals	range (s^{-1})	median	materially stabilizing ($\leq -\tau$)	materially destabilizing ($\geq \tau$)
A REAL	2770	-0.0069 to +0.0528	+0.0020	0	437
B REAL	2804	-0.0008 to +0.0570	+0.0101	0	1429

Replacing a synchronous machine by either converter never materially stabilizes the tracked electromechanical mode on this network, in any context. A reversal needs a stabilizing side of at least τ , so none can exist. F2 and F3 show the one-sided pattern.

11 Discrete curvature

- R8. Chain-identity verification is vacuous: no nested reversal exists at any level.
- Interaction terms still exist (F4). On the rightmost EM mode, second differences have both signs beyond $1e-3 s^{-1}$ in 13/16 A REAL policies and in 16/16 B policies. Their range is -0.0067 to $+0.0457 s^{-1}$ (A) and -0.0045 to $+0.0562 s^{-1}$ (B).
- Reference policy P68_10: 240 one-step squares with range -0.0052 to $+0.0009 s^{-1}$; 104 are below $-1e-3$ and none is above $+1e-3$. On the global $\alpha \perp$ all $|d| \leq 2.9e-5$.
- Reading. The set function is not modular, but its interactions do not change the sign of a replacement's effect: marginals shrink or grow with context and stay destabilizing.
- Post-hoc TDS pairs (§18). On their single-step chains the identity holds exactly (residual 0).

12 Fixed ranking

Source: results/CDW68_R09_gate.json.

stratum	Model A	Model B
FULL	57 decisions per policy (median), 0 material preferences: p^* undefined, regret 0	same
EM	0 decisions: not evaluable	same
LEVEL-D	0 decisions: not evaluable	same
TRACKED	51.5 decisions, $p^* = 1.0$, regret 0, transfer regret 0	46.5 decisions, $p^* = 1.0$, regret 0, transfer regret 0

- The insufficiency bar ($p^* \leq 0.90$ and $\text{regret} \geq 0.10$) is not met in any stratum.
- The T1 sign test cannot be computed.
- The headline "fixed ranking insufficient" is not allowed.
- On IEEE-68 the oracle fixed order of units reproduces every material preference on the tracked EM mode.

13 Total sensitivity validation (R11)

Source: results/CDW68_R12_gate.json.

model	test	n	sign agree- ment	median relative error	Spear- man	verdict
A	IFT total derivative (SPR-68) vs central re-equilibrated difference at $h = 1e-4$	80	1.000	$1.3e-5$	0.99995	PASS
B	numerical total derivative, $h = 1e-3$ vs $h = 1e-4$	80	1.000	$5.1e-7$	0.99991	PASS (numerical consistency has no analytical engine)

14 Reinforcement ranking (R12)

Holdout: 12 conditions per model, $\gamma = 1.5$.

	Model A	Model B
median Spearman(D_tot, finite)	0.9955	0.9810
selected static (discovery)	S4 electrical distance, ρ 0.197	S9 Δg SCR, ρ 0.434
median paired advantage (95 % bootstrap)	0.799 (0.797–0.800)	0.547 (0.546–0.547)
median top-5 precision	1.0	0.8
median Kendall / NDCG@5	0.958 / 1.000	0.896 / 0.998
ρ at $\gamma = 1.10 / 1.25 / 1.50$	0.9993 / 0.9980 / 0.9955	0.9984 / 0.9939 / 0.9810
T2 sign-flip p (Holm)	0.0078 (0.0156)	0.0898 (0.0898)
Wilcoxon p	0.00024	0.00024
gate	PASS	PASS

- T2B. It does not reach 0.05. The preregistered statistic is the median under exact sign-flip enumeration. The twelve B advantages are nearly equal in size, and in that case the median statistic has little power. The gate itself is defined on medians and passes.
- Materiality. The median over conditions of the largest finite effect is $4.2e-5 s^{-1}$ (A) and $3.3e-5 s^{-1}$ (B). In the median condition no branch has a material effect.
- Post-hoc EM-tracked ranking (A only). Median ρ 0.998 and top-5 1.0. The median of the per-condition largest effect is $0.020 s^{-1}$ (maximum 0.047). A median of 5 branches per condition exceed τ .
- Top branches.
 - EM-tracked, by median finite effect: lines 18–50, 51–50 and 18–49.
 - Preregistered target (both models): the G1 step-up transformer 1–54.

15 Static baselines

Discovery medians of Spearman with the finite $\times 1.5$ effect (results/CDW68_R10_static.csv):

index	A	B
S1 P	0.043	0.202
S2 S	0.047	0.221
S3 z	0.021	0.230
S4 electrical distance	0.194	0.337
S5 effective resistance	0.116	0.294
S6 Fiedler edge	0.093	0.242
S7 betweenness	0.029	0.081
S8 dV/dQ	0.123	−0.014
S9 Δg SCR	0.067	0.434

- The static indices do not change across conditions.
- In the holdout, the best static index per condition has median ρ 0.197 (A) and 0.434 (B).
- The Laplacian scores (S5, S6) are baselines only; no topology explanation is offered.

16 Governor/damping ablation (R13)

Model A, P68_01–06, 64 portfolios each.

variant	level D	level T	EM-tracked range (s^{-1})	stabilizing $\geq \tau$	destabilizing $\geq \tau$	largest global $ \Delta\alpha_{\perp} $
REAL	0/6	0/6	−0.0052 to +0.0393	0	267	1.5e-4

variant	level D	level T	EM-tracked range (s ⁻¹)	stabilizing ≥ τ	destabilizing ≥ τ	largest global Δα _⊥
NOGOV	0/6	0/6	-0.0102 to +0.0228	2	271	1.10e-2 (3 stabilizing, common-frequency mode, level B only)
SP33	0/6	0/6	-0.0136 to +0.0274	6	277	4.7e-3

- Removing governors, or both governors and damping, makes a few EM-tracked marginals materially stabilizing. None of them forms a nested reversal.
- Reversal is absent in the realistic model and in both simplifications. On this network the documented primary-frequency dynamics neither create nor remove it.
- The simplification does shift the EM marginals slightly toward stabilizing values, which is the direction the IEEE-39 limitation paragraph anticipated.

17 Cross-converter transfer

- What transfers between A and B:
 - the absence of reversal (0/16 in both);
 - one-sided destabilizing EM effects (B's are larger: median +0.0101 against +0.0020);
 - an error-free fixed ranking on the tracked mode;
 - the ranking method (R12 PASS in both).
- What does not transfer: the line list. Across matched conditions, the Spearman between the two models' finite truths has median 0.50 (range 0.00–0.51), Kendall 0.37 and top-5 overlap 0.4. The 0.00 is P68_16, where Model A's critical pole is a different pole.
- Which static index serves as comparator also depends on the converter: electrical distance for A, ΔgSCR for B.

18 Nonlinear TDS

Source: results/CDW68_R14_tds_posthoc.json, figure F10.

As preregistered. No case exists: there is no level-D or level-T reversal and no level-D-eligible negative control. R14 is "not applicable".

Post-hoc exploratory run. The run took the two EM-tracked nested pairs with the largest same-sign marginals at the reference policy P68_10.

- Model A REAL; 0.5 pu reactor at bus 3 for 10 s; decay of the tracked modal coordinate fitted over 11–29 s.
- Integrator: BDF with a network solve at each step.

pair	TDS decay change	linear decay change	sign agreement
G12, BASE → {11}	+0.011463 / +0.011467	+0.011460 / +0.011466	2/2
G4, {3} → {3,9}	+0.010275 / +0.010310	+0.010273 / +0.010307	2/2

The traces confirm the linear signs and magnitudes to within 4e-6 s⁻¹. They say nothing about reversals, because none exists.

19 Uncertainty

Deterministic stress envelopes E05 (±5 %) and E10 (±10 %) at the reference policy (results/CDW68_R15_uncertainty.json).

set	draws	level-D coverage	level-T coverage	draws with any materially stabilizing EM marginal	all 64 portfolios stable	smallest EM-tracked marginal
A E05	20	0/20	0/20	0/20	20/20	-0.0055
A E10	20	0/20	0/20	0/20	20/20	-0.0056
B E05	10	0/10	0/10	0/10	10/10	-0.0009

set	draws	level-D coverage	level-T coverage	draws with any materially stabilizing EM marginal	all 64 portfolios stable	smallest EM-tracked marginal
B E10	10	0/10	0/10	0/10	10/10	-0.0014

- Witness identity is undefined: the nominal reference has no witness and no draw creates one.
- Ranking under draws (Model A; first 5 draws of each envelope, 83 branches, $\times 1.5$): median ρ of D_{tot} 0.996 (10–90 % range over draws 0.9955–0.9960); post-hoc EM-tracked ρ 0.998 (0.9974–0.9981); median largest finite effect $4.3\text{e-}5 \text{ s}^{-1}$.

These are fractions of designed draws, not probabilities.

20 Cross-benchmark comparison

Source: results/CDW_CROSS_BENCHMARK_MATRIX.csv, figure F12.

	IEEE-39 A	IEEE-39 B	IEEE-68 A	IEEE-68 B
governors / damping	none / $D = 0$	none / $D = 0$	PST tg / PST d_o	PST tg / PST d_o
candidates, policies	9; 19 base-stable holdout	4; 7	6; 16	6; 16
level-D reversal	17/19 (median 0.0127 s^{-1})	global 0/7; EM-tracked 1/7	0/16 (T 0/16)	0/16 (T 0/16)
fixed-ranking regret	FULL 0.38; EM 0.03	not computed	FULL 0; TRACKED 0	FULL 0; TRACKED 0
total-sensitivity median ρ	0.996	0.993	0.996	0.981
static comparator median ρ	0.445	0.790	0.197	0.434
advantage	0.49	0.20	0.80	0.55
top-5	1.0	n/a	1.0	0.8
TDS	not performed	not performed	post hoc, signs 4/4	not applicable

21 Negative results

All are preserved in the result files.

1. The primary reversal gate fails on both models: 0/16 and 0/16.
2. Level T fails on both models: 0/16 and 0/16.
3. No materially stabilizing EM-tracked marginal exists in REAL for either model, in the census or in any of the 60 draws.
4. Ablations: 0/6 reversals without governors and 0/6 without governors or damping.
5. The fixed-ranking failure of IEEE-39 does not replicate. Every IEEE-68 stratum has regret 0 or cannot be evaluated.
6. The preregistered branch-ranking target is immaterial: its effects are $\leq 1.4\text{e-}3 \text{ s}^{-1}$, and in the median condition no branch is material.
7. T2B is not significant (sign-flip p 0.090).
8. The line list does not transfer between converter models (median ρ 0.50).
9. The preregistered TDS holdout has no case.
10. Entry 7 of the deviation log was wrong for NOGOV and is corrected in entry 11.

22 Novelty implications

See docs/CDW68_NOVELTY_ASSESSMENT.md.

- Nested contextual reversal and fixed-ranking failure become IEEE-39 benchmark-specific evidence.
- The one finding carried across two benchmarks and two converter models is the portfolio-conditioned total branch sensitivity used as a ranking. It is a known construction (Smed 1993; Nam et al. 2000; Joswig-Jones et al. 2026, preprint), validated against finite reinforcements on held-out policies and compared with discovery-selected static indices.
- On IEEE-68 its preregistered target is immaterial. The material EM version is post hoc.

- The word "first" is not used.

23 Paper decision

CASE B. The CDW manuscript (reports/papers/cdw_contextual_dynamic_weakness/) is re-titled and its contributions reordered:

1. Portfolio-conditioned re-equilibrated reinforcement ranking on two benchmarks and two converter models.
2. When re-equilibration matters: on IEEE-68 the frozen derivative reaches ρ 0.55 against 0.996; total and frozen disagree in sign for a median 33 % of branches.
3. Contextual reversal on IEEE-39, with its failure to replicate on IEEE-68.
4. Boundaries: fixed rankings, cross-model transfer, and the immaterial IEEE-68 target.

Readiness. The manuscript is not yet ready for TPWRS as a claims paper. The IEEE-68 ranking pass rests on an immaterial pole. Its material EM version is post hoc and needs a preregistered confirmation, or at least a second EM-target test on a fresh set of conditions.

24 Reproducibility

- Software: Python 3.13; NumPy, SciPy and pandas from .venv/tx3-analysis; ANDES 2.0.0 in .venv/xtool-andes-gf1. Thread counts are fixed to 1.
- Code in code/:

phase	scripts
inputs and audit	R00_inputs.py, R01_audit.py, R01b_manifest.py
Model B	R02_export_b.py, andes68/run68.py
census and reversal	R06_census.py, R07_reversal.py
fixed ranking	R09_ranking.py
branches and gates	R10_branches.py, R10_static.py, R12_ranking_gate.py
TDS, uncertainty, synthesis	R14_tds.py, R15_uncertainty.py, R18_crossbench.py, R21_figures.py

- Raw per-task JSON in raw/ is not committed. It is listed with sha256 in CDW68_RAW_MANIFEST.csv and packed in CDW68_RAW_RESULTS_20260913.zip.
- Runtimes (wall clock):

phase	time
census, 1792 cases	451 s
Model B census	8 shards, 55 s
Model B branches	6 shards, $\leq$ 919 s
Model B draws	5 shards, $\leq$ 160 s
Model A branches, 128 tasks	1862 s
Model A draws, 2560 cases	703 s
ranking under draws, 40 tasks	712 s

- The campaign ran from the preregistration commit (09:19) through the last computation (R15L, finished 10:48:02) on 2026-09-13; the times of the document, paper and bundle commits are in git.
- Figures: F1–F12 in figures/ (PDF, SVG, PNG).